

$\chi, \deg, \text{rk}$ of coherent sheaves

Let X be an irr proj. curve, $\mathcal{F}$ a coherent sheaf. Let

$$\chi(\mathcal{F}) = h^0(\mathcal{F}) - h^1(\mathcal{F}),$$

which is additive on s.e.s. We call $\mathcal{F}$ a torsion sheaf if its support is 0-dimensional, ie if its stalk at the generic point is 0. Then $h^1(\mathcal{F}) = 0$ (eg by Grothendieck's theorem).

Note that also in this case

$$\# \text{Supp } \mathcal{F} \leq h^0(\mathcal{F}).$$

Since X is 1-dimensional,

$$\chi(\mathcal{F}(n)) = a + bn$$

for some a, b . We define

$$\begin{aligned} \deg \mathcal{F} &= \chi(\mathcal{F}) - \text{rk } \mathcal{F} \chi(\mathcal{O}_X) \\ &= \chi(\mathcal{F}) - \text{rk } \mathcal{F} (1-g), \end{aligned}$$

which is also additive.

Note: If a line bundle $\mathcal{L}$ has a section, we get a s.e.s.

$$0 \rightarrow \mathcal{O}_X \rightarrow \mathcal{L} \rightarrow T \rightarrow 0,$$

where T is torsion. Hence,

$$\deg \mathcal{L} = \chi(\mathcal{L}) - \chi(\mathcal{O}_X) = \chi(T) = h^0(T) > 0.$$

We also have

$$\deg \mathcal{F} \otimes \mathcal{L} = \deg \mathcal{F} + \deg \mathcal{L} \text{rk } \mathcal{F}.$$

In particular,

$$\chi(\mathcal{F}(n)) = \deg \mathcal{F} + \text{rk } \mathcal{F} (n \cdot \deg \mathcal{H} - \chi(\mathcal{O}_X)).$$

We say that $h: G \rightarrow F$ is generically $\square$ if the induced maps $h_x: G_x \rightarrow F_x$ are $\square$ for almost every $x \in X$.

Note: If $h: G \rightarrow F$ is gen. an isomorphism then

- h is injective,
- $F/h(G)$ is torsion.

In particular,

$$\begin{aligned} \chi(G) &= \chi(F) - \chi(F/h(G)) \\ &= \chi(F) - h^0(F/h(G)) \leq \chi(F). \end{aligned}$$

Hence, $\deg G \leq \deg F$.

Now let X be non-singular. Let $G \subseteq F$ be vector bundles. Let G' be the inverse image of the torsion subsheaf of F/G . Then

- $\text{rk } G = \text{rk } G'$,
- $\deg G' \geq \deg G$. From $0 \rightarrow G \rightarrow G' \rightarrow G'/G \rightarrow 0$, G'/G is torsion.

We call G' the vector subbundle generically generated by G . Note that F/G' is locally free.

Note:

- $F \subseteq \mathcal{O}_X \Rightarrow \deg F \leq 0$ Subsheaves of locally free sheaves are locally free, so either $F=0$ or $\deg F = \chi(F) - \chi(\mathcal{O}_X) = \chi(\mathcal{O}_X/F) = -h^0(\mathcal{O}_X/F)$.
- if F is generically generated by global sections then $\deg F \geq 0$
From $0 \rightarrow K \rightarrow \mathcal{O}_X^n \rightarrow F \rightarrow T \rightarrow 0$, we see that $\text{rk } F = \text{rk } \mathcal{O}_X^n/K$ and $\deg F \geq \deg \mathcal{O}_X^n/K$, so it suffices to show $\deg \mathcal{O}_X^n/K \geq 0$, ie that $\deg K \leq 0$.
- a locally free sheaf F has a rank 1 subbundle
Take a section of $F \otimes \mathcal{H}^*$ to get a map $\mathcal{O} \rightarrow F \otimes \mathcal{H}^*$. Take the line bundle generically generated by the image as subbundle.

Now let X be a non-singular projective curve. Let $\mu(F) = \frac{\deg F}{\text{rk } F}$. Then F is (semi-) stable if for any subbundle $0 \neq G \neq F$, $\mu(G) \leq \mu(F)$.

Then line bundles are stable, and twisting by line bundles does not change (semi)stability.

Lemma: If F_1, F_2 are stable with $\mu(F_1) = \mu(F_2)$ then any non-zero homomorphism $h: F_1 \rightarrow F_2$ is an isomorphism.

Let $G_1 \subseteq F_1, G_2 \subseteq F_2$ be gen. generated by $\text{Ker } h, \text{Im } h$. Then

$$\begin{aligned} \text{rk } F_1 &= \text{rk } \text{Ker } h + \text{rk } \text{Im } h = \text{rk } G_1 + \text{rk } G_2, \\ \deg F_1 &= \deg \text{Ker } h + \deg \text{Im } h \\ &\leq \deg G_1 + \deg G_2 \quad \text{This is an equality iff } \text{Im } h = G_2, \\ &= \text{rk } G_1 \mu(G_1) + \text{rk } G_2 \mu(G_2) \quad \text{Ker } h = G_1 \\ &\leq \text{rk } G_1 \mu(F_1) + \text{rk } G_2 \mu(F_2) \quad \text{This is an equality iff } \mu(G_1) = \mu(F_1), \text{ ie } \\ &= (\text{rk } G_1 + \text{rk } G_2) \mu(F_1) = \deg F_1. \quad \text{iff } G_1 = \mathcal{O}_1 G_2 = F_2. \end{aligned}$$

Corollary: Stable bundles are simple (ie $\text{End} = \text{Aut}$).

If $h: F \rightarrow F$ is not 0 then $h_x: F_x \rightarrow F_x$ has a non-0 vector, so $h^{-1}x$ is not an isomorphism, hence 0.

Lemma: Let F be semistable of rank r , degree d , with $d \geq r(2g-1)$. Then

- F is generated by global sections,
- $H^1(F) = 0$.

We have $H^1(F) = H^0(K \otimes F^*)$. Let $h: F \rightarrow K$. Let G be generated by $\text{Ker } h$. Then $\text{rk } G = r-1$ if $G \neq F$, ie if $h \neq 0$.

$$\deg G \geq \deg \text{Ker } h \geq \deg F - \deg K = d - 2g + 2.$$

By semistability, $\mu(G) \leq \mu(F)$, ie $\frac{\deg G}{r-1} \leq \frac{d}{r}$, ie $r \deg G \leq (r-1)d$.

In particular,

$$\begin{aligned} r(d-2g+2) &\leq (r-1)d, \text{ ie} \\ r d - 2rg + 2r &\leq r d - d, \text{ ie} \\ d &\leq 2r(g-1), \end{aligned}$$

unless $h=0$.

$$0 \rightarrow \text{Im } h \rightarrow F \rightarrow E \rightarrow 0$$

For the first statement, we want to show that $H^0(F) \rightarrow F_x$, ie that $H^1(\text{Im } h) = 0$.

Note that $\text{Im } h = \mathcal{O}(x)$, so $F \otimes \text{Im } h$ is semistable. Also,

$$0 \rightarrow \text{Im } h \rightarrow \mathcal{O}_x \rightarrow k \rightarrow 0$$

gives $\deg \text{Im } h = -1$. Hence, $\deg F \otimes \text{Im } h = d - r > r(2g-2)$. Now apply the previous part.

Corollary: Let $g=0$, $\mathcal{H}$ a line bundle of degree 1. Then

- any line bundle is of the form $\mathcal{H}^d$ for some d ;
- X has no stable bundles of rank $r > 1$;
- its semistable bundles of rank r are of the form $\mathcal{O}_X^{\oplus s} \otimes \mathcal{H}^{\otimes m}$.

- Let $L \otimes \mathcal{H}^{-d}$ have degree 0. Then it is generated by its sections, so we get

$$\mathcal{O}_X \rightarrow L \otimes \mathcal{H}^{-d},$$

a generic isomorphism, hence an isomorphism (both are stable of rank 1).

- Let F be stable of rank $r > 1$. Let m be s.t.

$$-r < \deg(F \otimes \mathcal{H}^{\otimes m}) \leq 0.$$

Then $F \otimes \mathcal{H}^{\otimes m}$ has a section and hence a line subbundle of non-negative degree.

As above, we get $\deg(F \otimes \mathcal{H}^{\otimes m}) = 0$. Choose r sections which are linearly independent at $x \in X$. We get a generic isomorphism $\mathcal{O}_X^{\oplus r} \rightarrow F \otimes \mathcal{H}^{\otimes m}$. Hence, it is injective and the quotient is torsion. Since $\deg \mathcal{O}_X^{\oplus r} = 0 = \deg(F \otimes \mathcal{H}^{\otimes m})$, it is an isomorphism.